Student number:	Seat No:

Mid-Semester Test – Sm1 2026

FNCE30007
Writing Time: 60 minutes
Reading Time: None
Weighting: 25%

Permitted Materials

- Calculator Casio FX82 (any suffix)
- Mathematical instruments (Protractor, compass, ruler, circle template).
- Formula sheet provided.

Instructions to Students

- During writing time, clearly write your student number and seat number on the cover sheet in the above boxes only. Ensure each digit is clearly written.
- Please write the answers in CAPITAL LETTERS only in the designated answer boxes provided within the exam paper for each question.
- Use a pen and not a pencil to write your answers
- Do not communicate with other students in the exam venue.
- No part of the exam paper may be removed from the venue.
- Ensure you only have standard permitted materials on your desk plus additional materials if listed above.
- Raise your hand if you require assistance.
- Students are NOT permitted to take pictures or make any reproduction via any means of any of the questions in the exam paper.

- Test contains 13 multiple-choice questions.
- Total marks: 15.
- Questions 5 and 6 are worth 2 marks each. All other questions are worth 1 mark.
- Select one answer to each question.

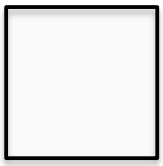
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Question 1 (1 mark)

A cereal manufacturer needs wheat as input in their production, hedges by taking a long position in wheat futures and needs to close out the hedge before the expiration of the futures contracts. Just before the hedge is closed out, however, the basis (spot - futures) unexpectedly weakens. Relative to what had been expected prior to the weakening of the basis, which of the following is correct?

A. The effective price paid is higher than expected.
B. The effective price paid is unchanged because futures eliminate basis risk.
C. The effective price paid by the manufacturer is lower than expected.
D. The hedge becomes a perfect hedge.
E. One cannot tell unless the contract is cash settled.

Write your CAPITAL LETTER answer in the box

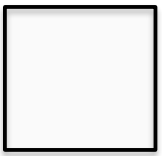


Question 2 (1 mark)

An airline hedges long-dated jet-fuel needs with short-dated gasoline futures that it plans to roll forward repeatedly. Even if the hedge is sensible economically, which problems can the hedger face due to the daily settlement of futures?

A. Elimination of basis risk and, thus, elimination of large return opportunities
B. Arbitrage losses
C. Counterparty risk, as the clearinghouse may not absorb all losses
D. Liquidity risk from margin calls and rollover losses
E. Quantity risk, as it may not receive gasoline deliveries

Write your CAPITAL LETTER answer in the box

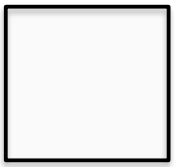


Question 3 (1 mark)

An FT article discussed in class described a sharp decline in futures hedging by chocolate makers. Which explanation (provided in the article) best explains such decline?

A. The fact that long futures expose hedgers only to upside
B. Higher initial-margin or cash-financing needs and a view that prices could fall from extreme levels
C. The fact that cocoa had become an investment asset
D. Lower cocoa prices and a rising convenience yield
E. Greater concern about price risk than quantity risk

Write your CAPITAL LETTER answer in the box

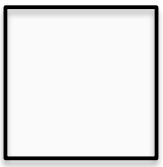


Question 4 (1 mark)

When shortage risk rises and the oil futures curve (and, in particular, the prompt-deferred oil price spread) steepens sharply, the most likely interpretation is:

A. A guaranteed positive return from going long prompt oil and short deferred oil
B. Lower convenience yield and deeper contango
C. Zero basis at all maturities
D. Lower spot prices relative to futures prices
E. Higher convenience yield and deeper backwardation

Write your CAPITAL LETTER answer in the box

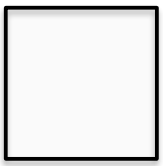


Question 5 (2 marks)

An actively managed equity portfolio is worth $21,000,000 and currently has a beta of 0.70. The portfolio manager wants to raise the beta to 1.30 for the next four months using S&P500 futures. The futures contract the manager wants to rely on is trading at 4,200, and each contract is $50 times the index. Ignoring any small maturity mismatch, which is the most appropriate futures trade?

A. Short 60 contracts
B. Long 42 contracts
C. Long 60 contracts
D. Long 100 contracts
E. Short 100 contracts

Write your CAPITAL LETTER answer in the box

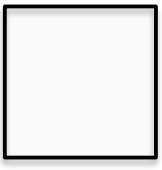


Question 6 (2 marks)

A 6-month futures contract is written on gold, an investment commodity. The spot price is $1,200 per ounce. The storage cost over the next 6 months is $8 per ounce, payable at the end of the 6 months. The risk-free rate is 2% per annum with continuous compounding. The 6-month futures price is quoted at $1,226.12 per ounce. Ignoring transaction costs, which statement is correct?

A. Go short futures and buy gold spot financed by borrowing, earning about $6.06 per ounce.
B. Go long futures and short gold spot, earning about $6.06 per ounce.
C. There is no arbitrage because the futures price equals the no-arbitrage price.
D. Go short futures and buy gold spot financed by borrowing, earning about $14.06 per ounce.
E. There is no arbitrage because gold pays no income.

Write your CAPITAL LETTER answer in the box

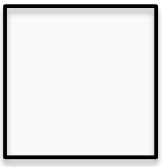


Question 7 (1 mark)

A trader goes short 6 corn futures contracts at $4.30 per bushel. Each contract is on 5,000 bushels. The initial margin is $2,000 per contract and the maintenance margin is $1,600 per contract. At the end of the first trading day, the settlement price is $4.22 per bushel. Which statement is correct?

A. The balance rises to $16,800, so up to $4,800 may be withdrawn.
B. The balance rises to $14,400, so up to $2,400 may be withdrawn.
C. The balance is unchanged because futures gains are realized only at maturity.
D. The balance falls to $9,600 and a margin call of $2,400 is required.
E. The balance rises to $12,480, but nothing may be withdrawn until the contract expires.

Write your CAPITAL LETTER answer in the box

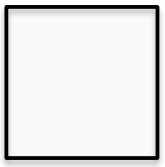


Question 8 (1 mark)

Which statement is most consistent with the lecture discussion on arguments for and against corporate hedging?

A. If competitors do not hedge, hedging is guaranteed to reduce risk.
B. Hedging is successful only if both the hedge and the underlying asset make gains.
C. Because shareholders are diversified, hedging can never reduce the risk of the firm.
D. One argument in favor is that firms should focus on their core business rather than take views on market variables; one argument against is that diversified shareholders may choose their own hedges.
E. Hedging is justified only if managers can forecast commodity, FX, or interest-rate movements better than the market.

Write your CAPITAL LETTER answer in the box

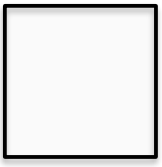


Question 9 (1 mark)

The current spot price of wheat is $70. The 6-month futures price is $68. Analysts expect the spot price in 6 months to be $69. Which statement is correct?

A. The market is in “industry” contango and normal contango.
B. The market is in “industry” contango and normal backwardation.
C. The market is in “industry” backwardation and normal contango.
D. One cannot classify the market without knowing interest rates.
E. The market is in “industry” backwardation and normal backwardation.

Write your CAPITAL LETTER answer in the box

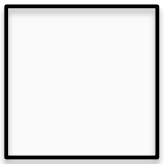


Question 10 (1 mark)

An Australian exporter will receive EUR 6 million in four months. To lock in the AUD value using the forward market, the exporter should:

A. go short AUD forwards because it is naturally short EUR.
B. go short EUR forwards because it is naturally long EUR.
C. do nothing because forwards can hedge only known foreign-currency outflows.
D. buy EUR spot today and finance it by borrowing AUD.
E. go long EUR forwards because it is naturally long EUR.

Write your CAPITAL LETTER answer in the box

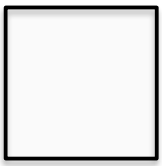


Question 11 (1 mark)

A European put option on a stock has current stock price $47, strike price $50, and premium $5.20. Which statement is correct?

A. Intrinsic value = $3.00; time value = $2.20
B. Intrinsic value = $2.20; time value = $3.00
C. Intrinsic value = $50.00; time value is negative
D. Intrinsic value = $5.20; time value = $0.00
E. Intrinsic value = $3.00; time value = $5.20

Write your CAPITAL LETTER answer in the box

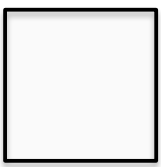


Question 12 (1 mark)

Which statement is correct for a European put option on a non-dividend-paying stock with strike K and maturity T?

A. Its premium falls when volatility rises.
B. Its value can exceed K e^(-rT) if it is in the money.
C. It is always worth more than the corresponding American put.
D. Its value can never exceed K e^(-rT).
E. It must be worth zero whenever S < K.

Write your CAPITAL LETTER answer in the box



Question 13 (1 mark)

The Financial Times article "Wall Street turns to complex trades to dodge AI ‘implosions’" described dispersion trades, in which investors buy single-stock volatility and sell index volatility. In which market environment is such a trade most likely to perform well?

A. When the S&P 500 and most stocks fall together in a broad market sell-off.

B. When index volatility rises more than single-stock volatility.

C. When individual stocks experience large idiosyncratic swings while the index remains relatively stable.

D. When interest rates are constant and futures prices equal forward prices.

E. When AI-related stocks all rise by about the same percentage.

Write your CAPITAL LETTER answer in the box

